

Computer Science & IT

Theory of Computation



Context Free Languages

Lecture No. 1



By- Ravindra Sir

Recap of Previous Lecture



Topic

RL

Topic

Topic

Topics to be Covered



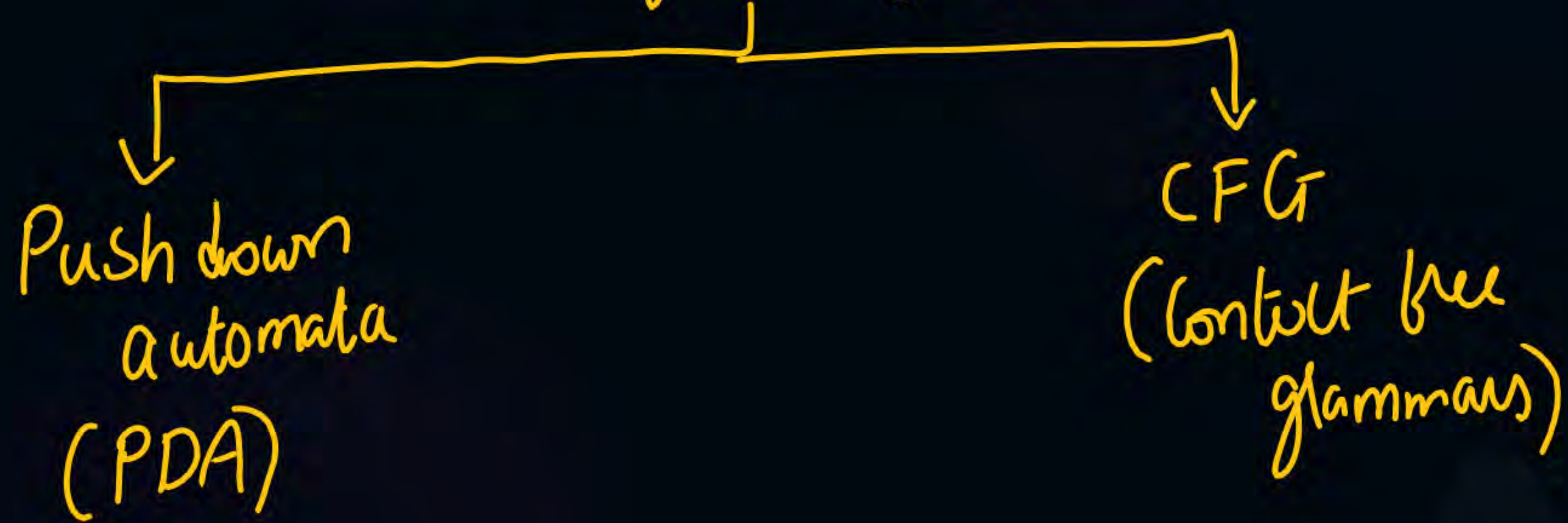
Topic

CFL

Topic

Topic

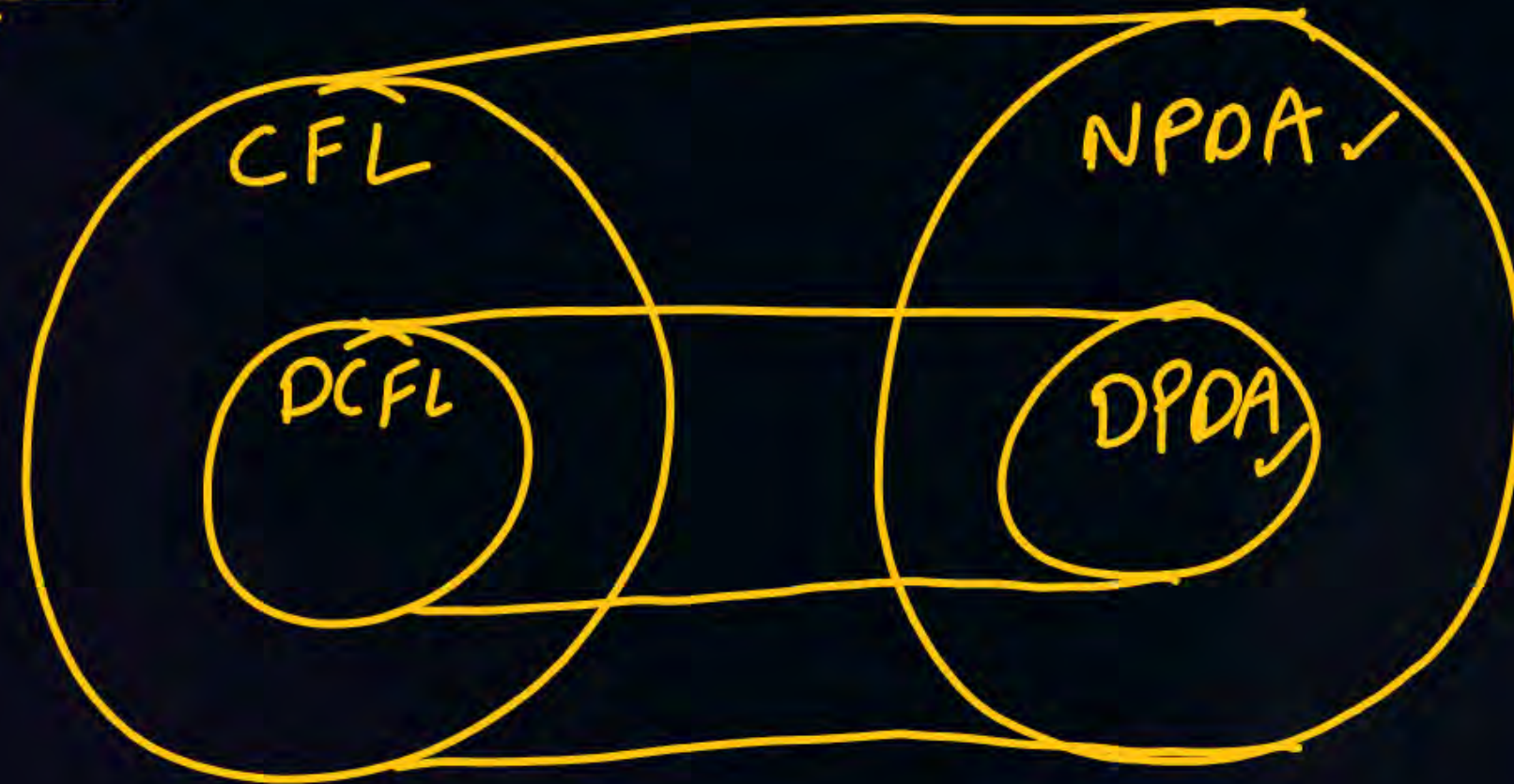
Context free languages



$NFA \cong DFA$

DPDA $<$ NPDA ✓

DPDA ✓
NPDA

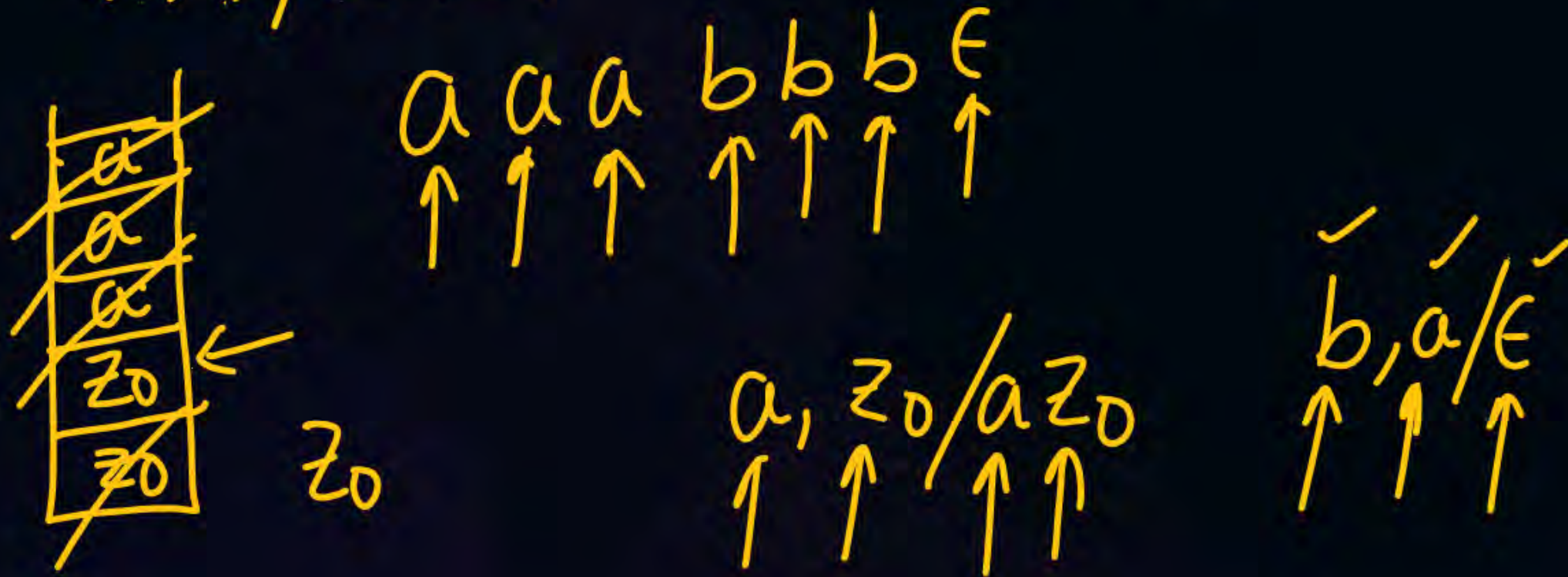
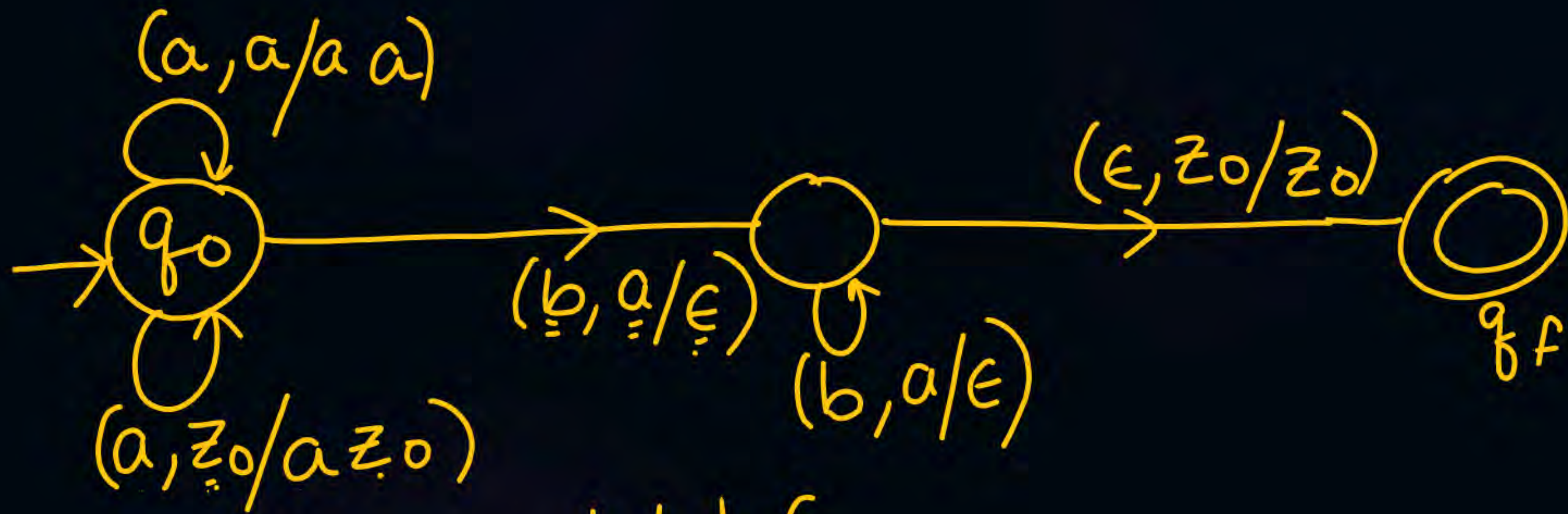


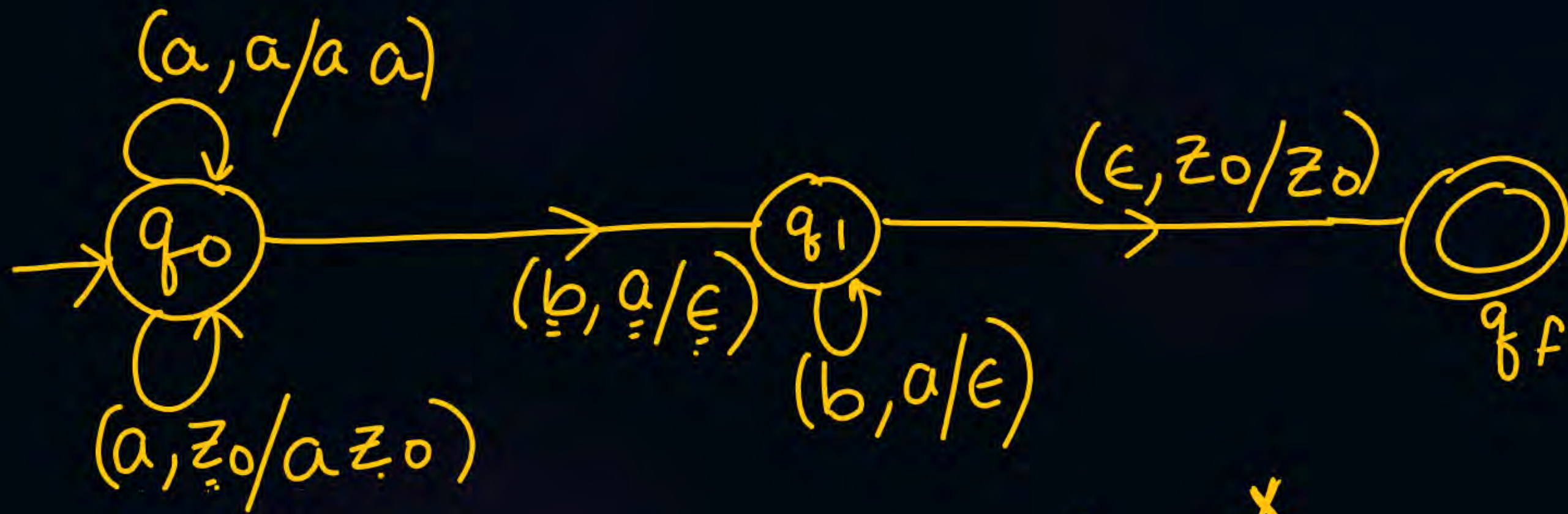
PDA = FA + Stack



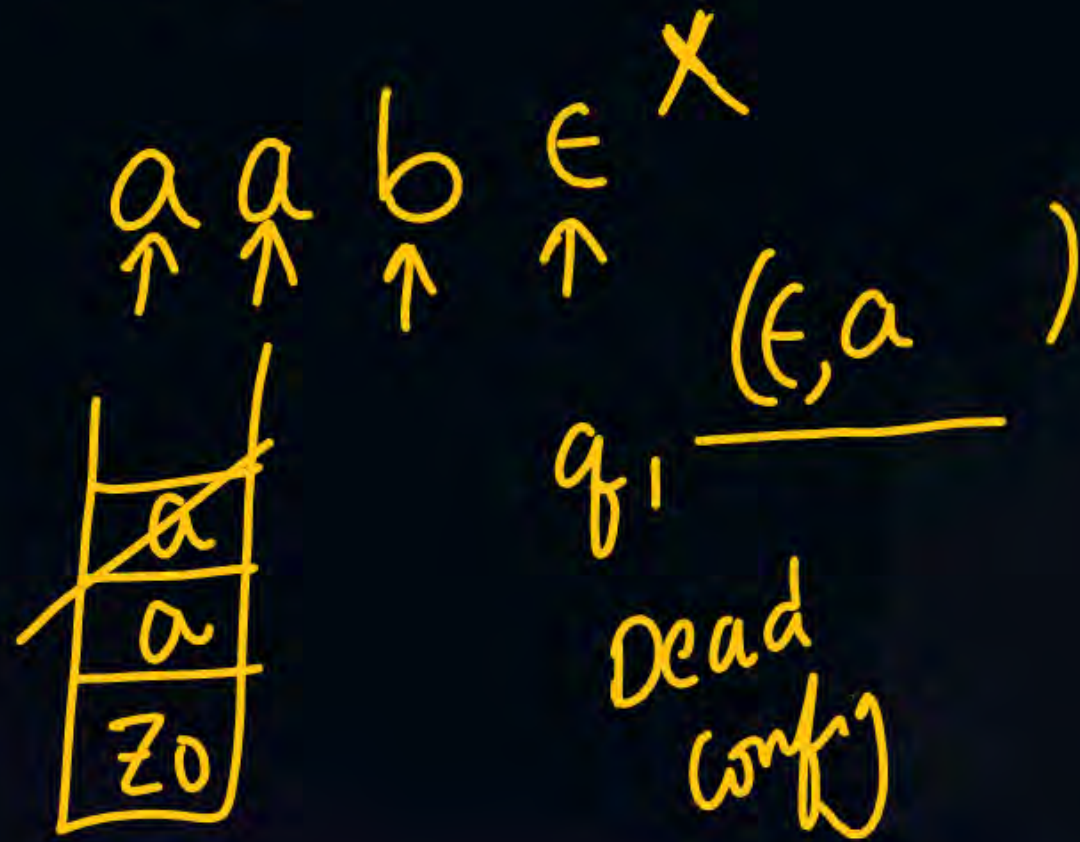
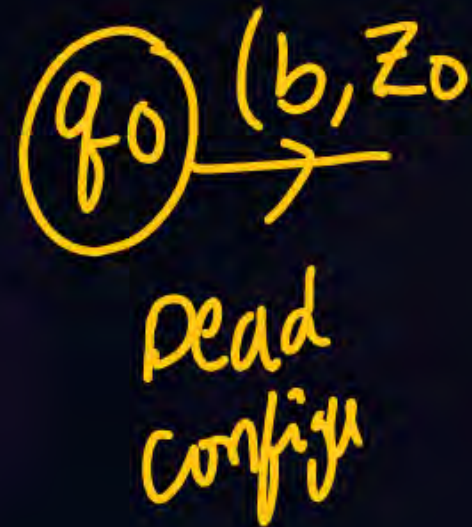
2 min time

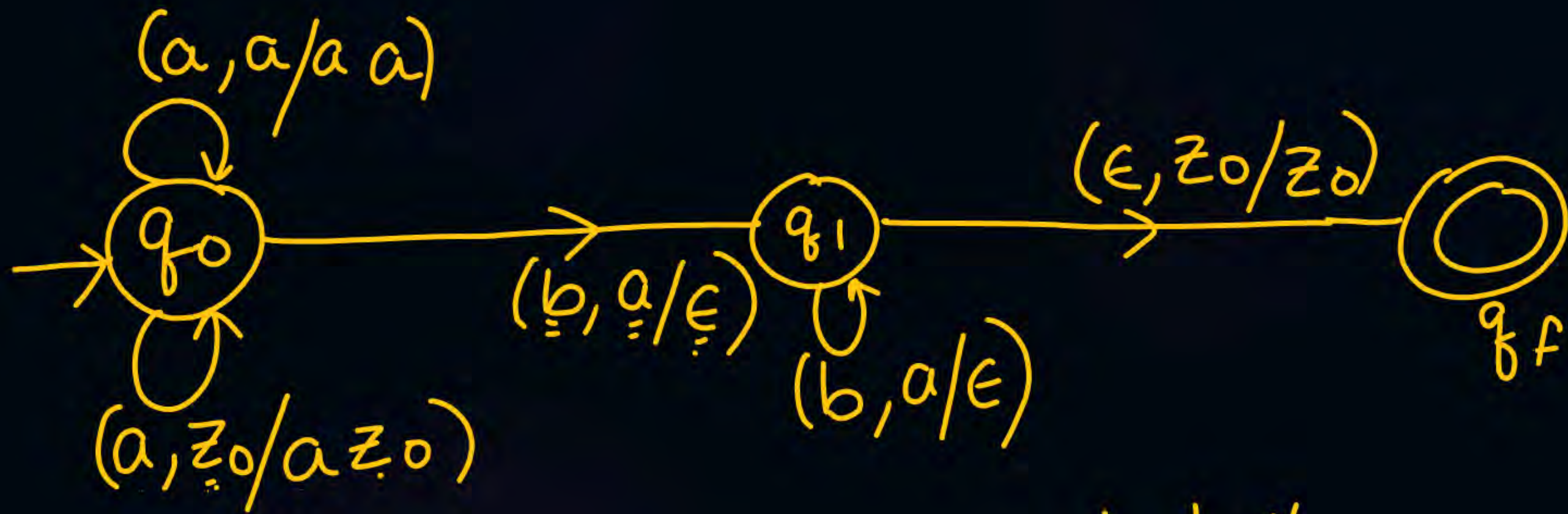
in PDA, there is no need to minimize the no of states
2 min



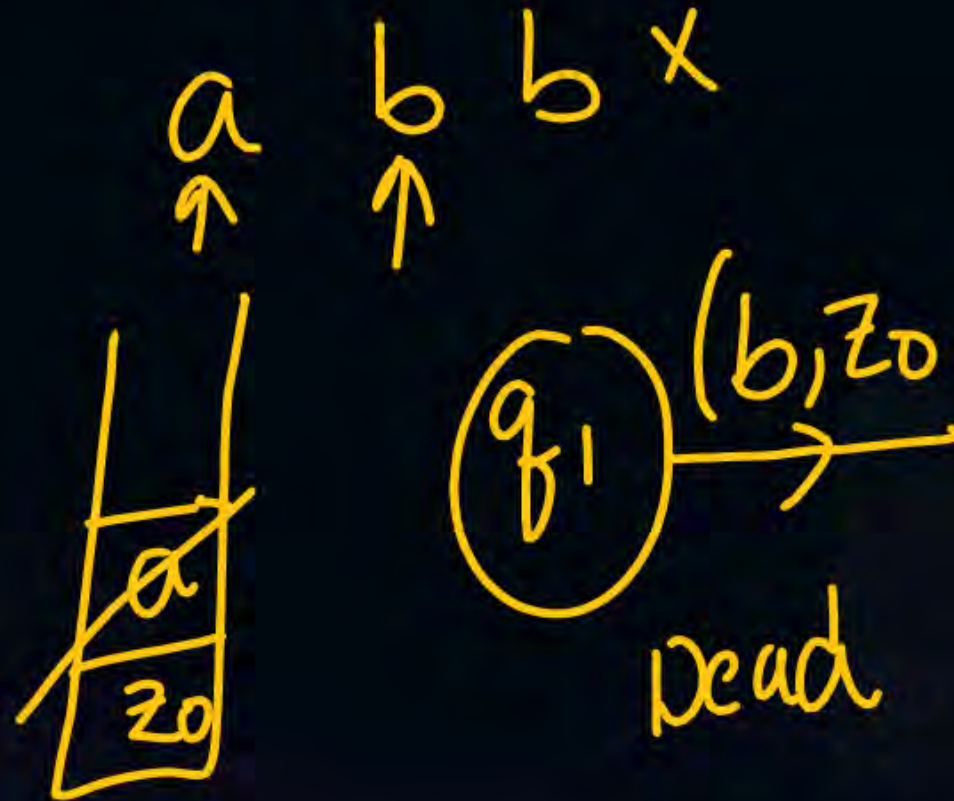
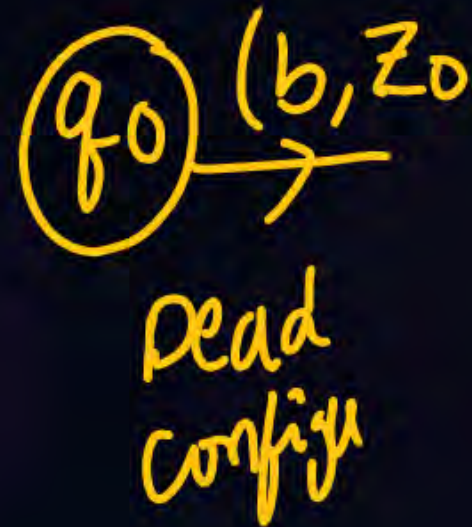


$b a a^x$





$b \ a \ a^x$



$$\omega / n_a(\omega) = n_b(\omega)$$

aa $\begin{matrix} \uparrow & \uparrow \\ b & b \end{matrix}$



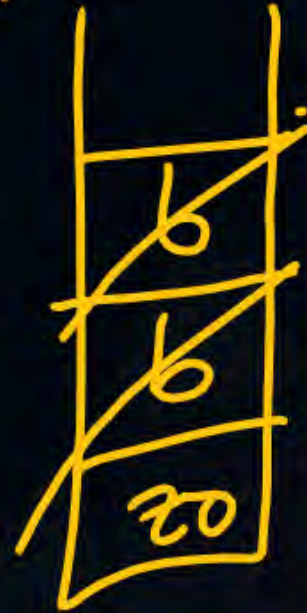
a $\begin{matrix} \uparrow & \uparrow & \uparrow \\ b & a & b \end{matrix}$



b $\begin{matrix} \uparrow & \uparrow & \uparrow \\ b & a & a \end{matrix}$



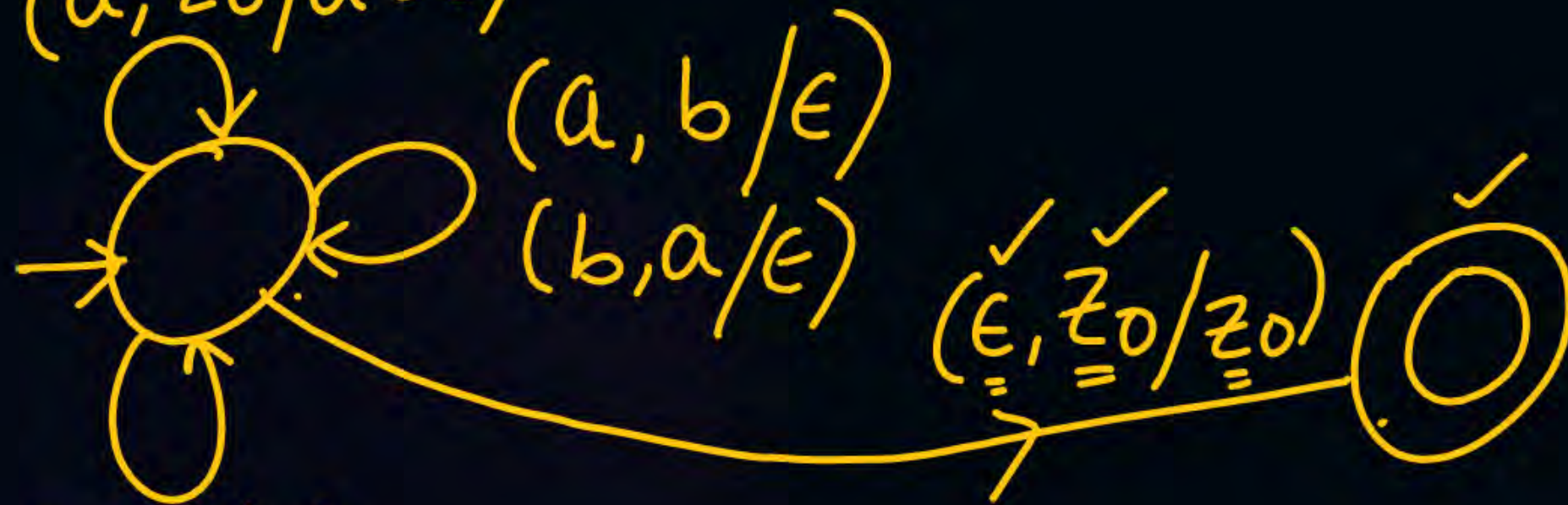
b $\begin{matrix} \uparrow & \uparrow & \uparrow & \uparrow \\ a & b & a \end{matrix}$



$$w/n_a(w) = n_b(w) /$$

$$(b, z_0 / bz_0)$$

$$(a, z_0 / az_0)$$

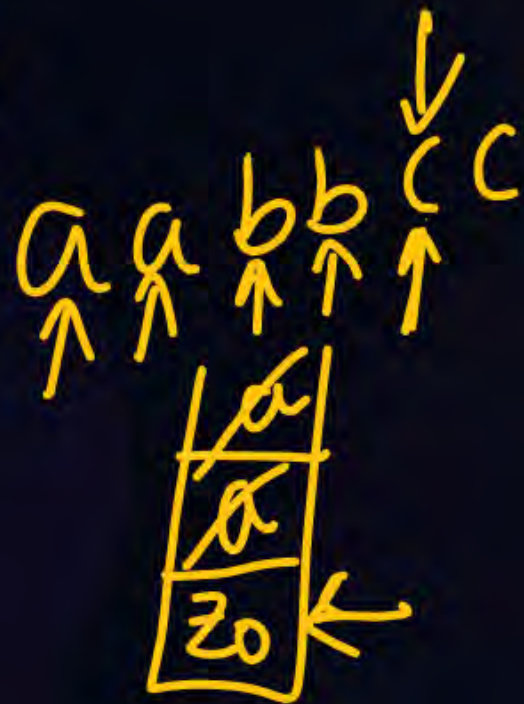
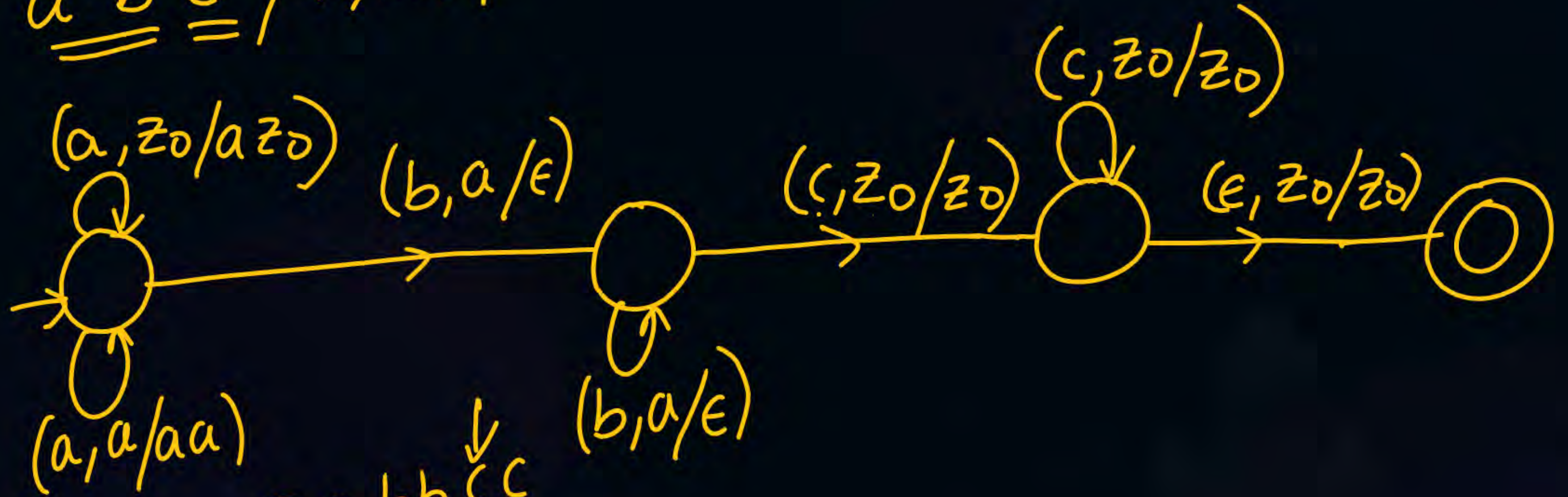


$$(a, a/aa)$$

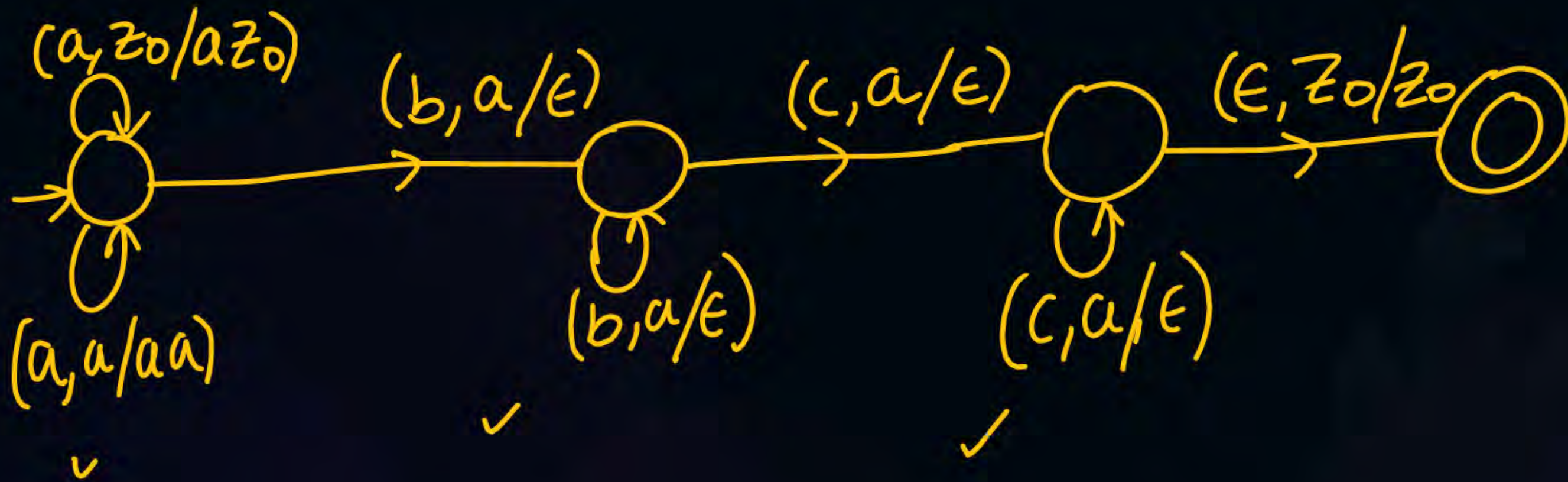
$$(b, b/bb)$$

DPDA ✓

$$\underline{a^n b^n c^m} / n, m \geq 1$$



$$\underline{a}^{n+m} \underline{b}^m \underline{c}^n / m, n \geq 1$$



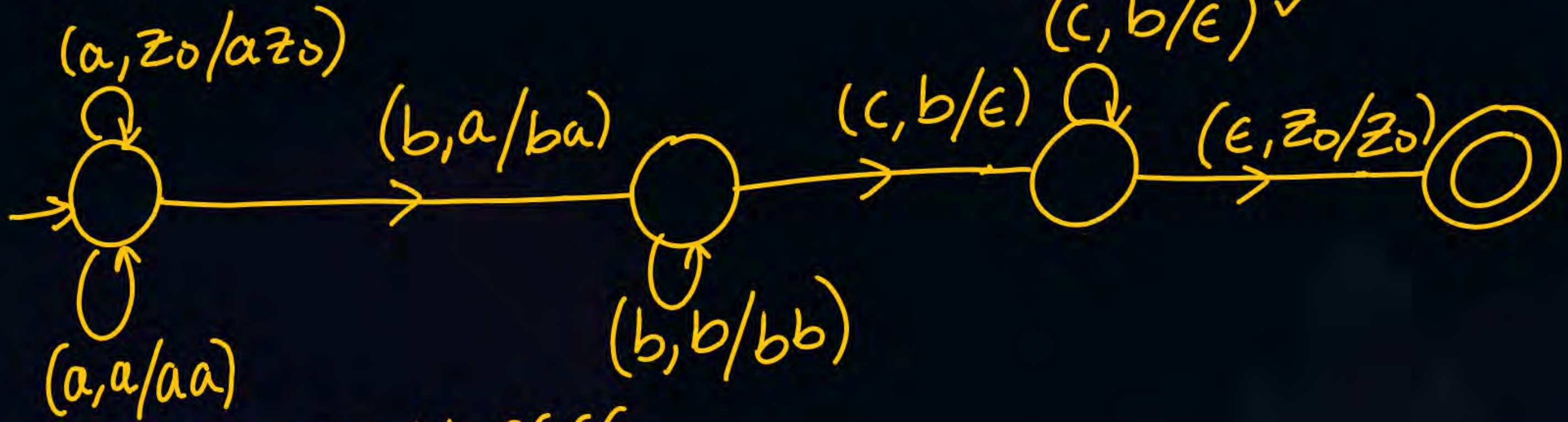
$$a^n b^{m+n} c^m / m, n \geq 1$$

$$\underline{a}^n \underline{b}^n \underline{b}^m \underline{c}^m / m, n \geq 1 \quad \begin{matrix} (b, \underline{b}/bb) \\ (b, \underline{z_0}/bz_0) \end{matrix}$$



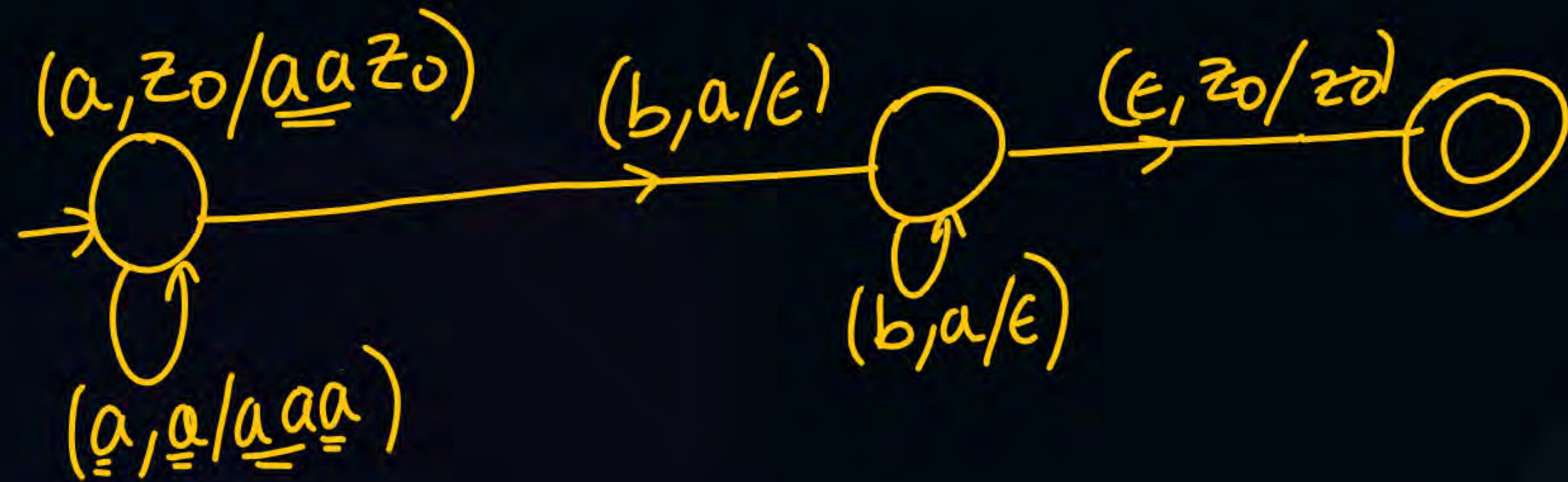
aabb(bb)cc

$$\underline{a^n b^m c^{n+m}} / n, m \geq 1$$

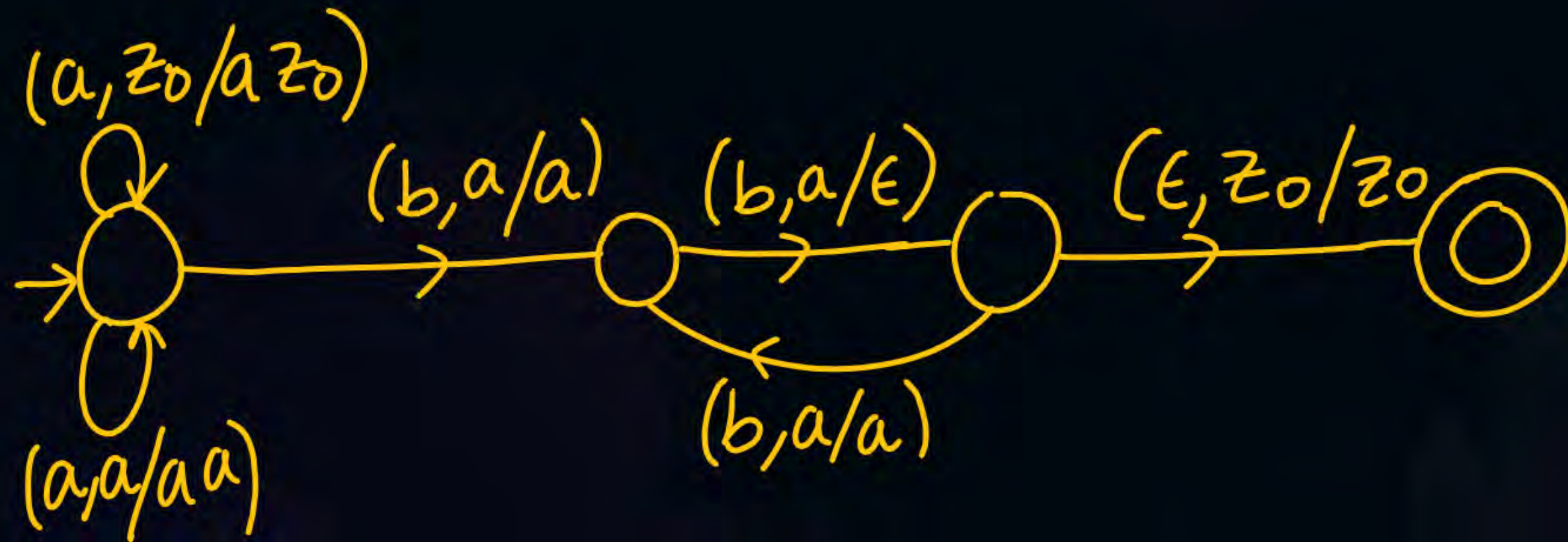


$aa\cancel{bb}ccc$
 $\uparrow$
 $\boxed{\cancel{a}\cancel{a}\cancel{b}\cancel{b}}$

$$a^n b^{2n} / n \geq 1$$



$a^n b^{2n} / n \geq 1$ (push 1 a for every a and for every 2 b's, pop a 'a')



aa bbbbbb(b)b
 ↑↑↑↑
 a
 a
 a
 a
 20 ←

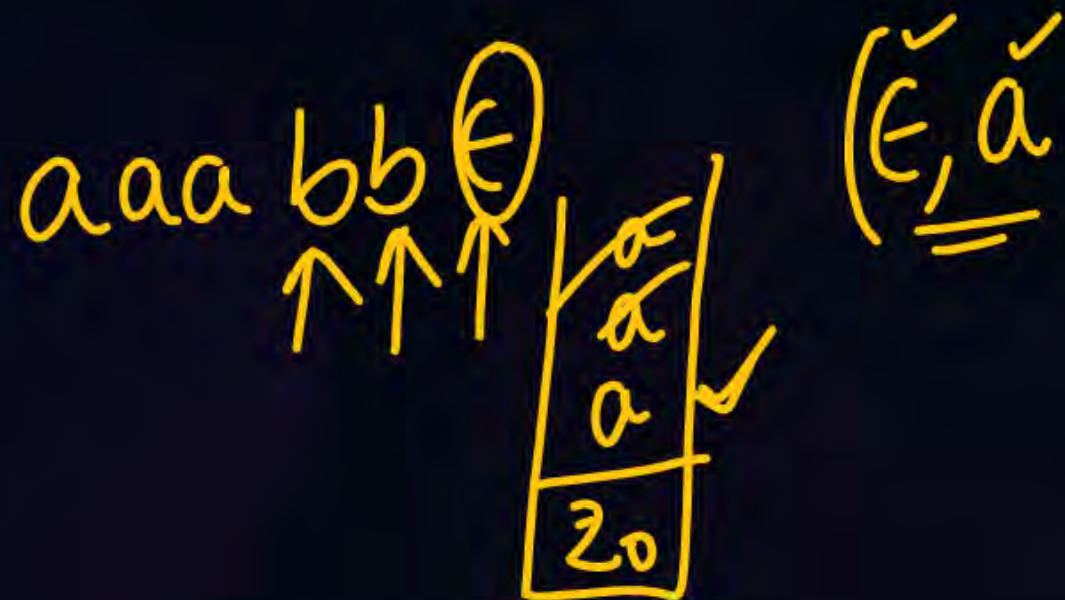
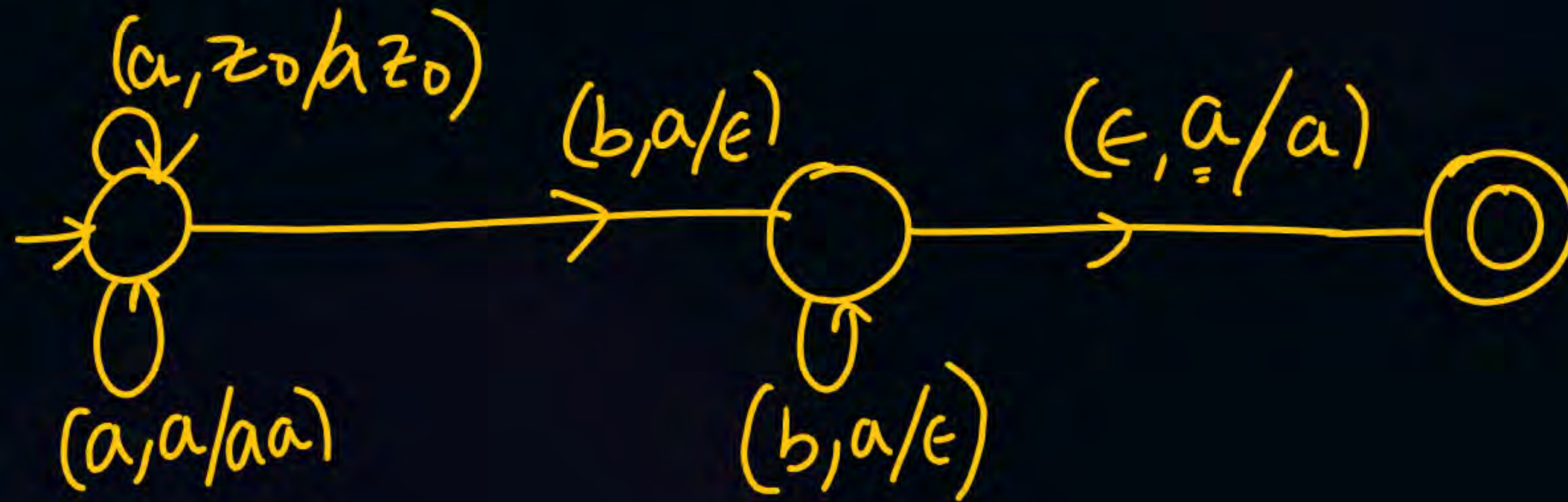
$\overbrace{a^n} \overbrace{b^n} \overbrace{c^n} / n \geq 1 \quad X \rightarrow \text{CSL}$
 $\downarrow \quad \downarrow$
 $\textcircled{2a's} \quad \textcircled{\text{pop a's}} \quad \textcircled{XCFL} \checkmark$

aaa bbcccc ✓

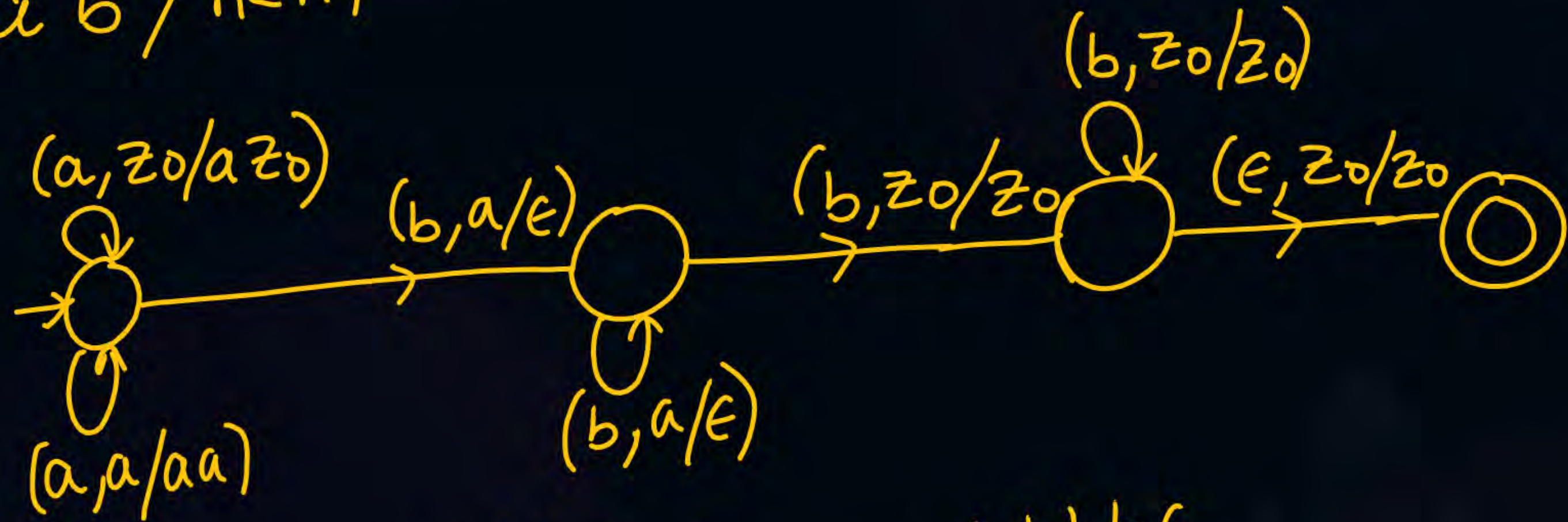
$\begin{array}{|c|} \hline \cancel{a} \\ \hline \cancel{a} \\ \hline \cancel{a} \\ \hline \cancel{a} \\ \hline \cancel{a} \\ \hline \cancel{a} \\ \hline \end{array}$

$a^m b^n c^l / \underline{\underline{2m = n + l}} \checkmark$

$a^n b^m / n > m, n, m \geq 1$



$$a^n b^m / n < m$$



aa $\uparrow$ $\uparrow$ $\uparrow$ b b b b ϵ

a
a
z0

a a b b b b ϵ

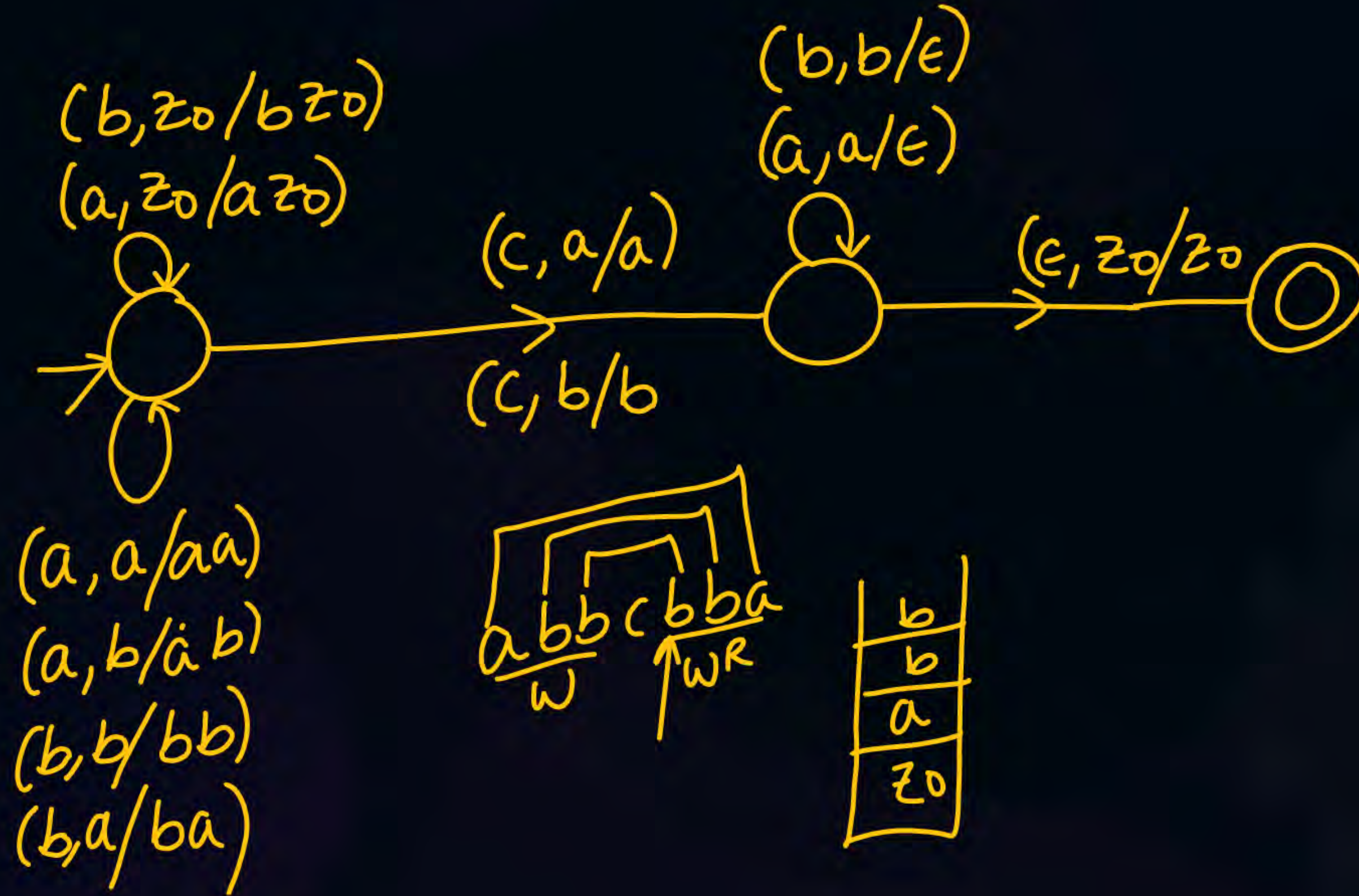
$wcw^R / w \in (a+b)^+$, c is a special symbol
palindrome

$w = abb$

$wcw^R = \underline{abb} \overset{\checkmark}{c} \underline{bba}$
 ↑↑↑

b
b
a
ε

DPDA → Centre is known





Till now all the PDA's we have seen are DPDA
↓
deterministic PDA

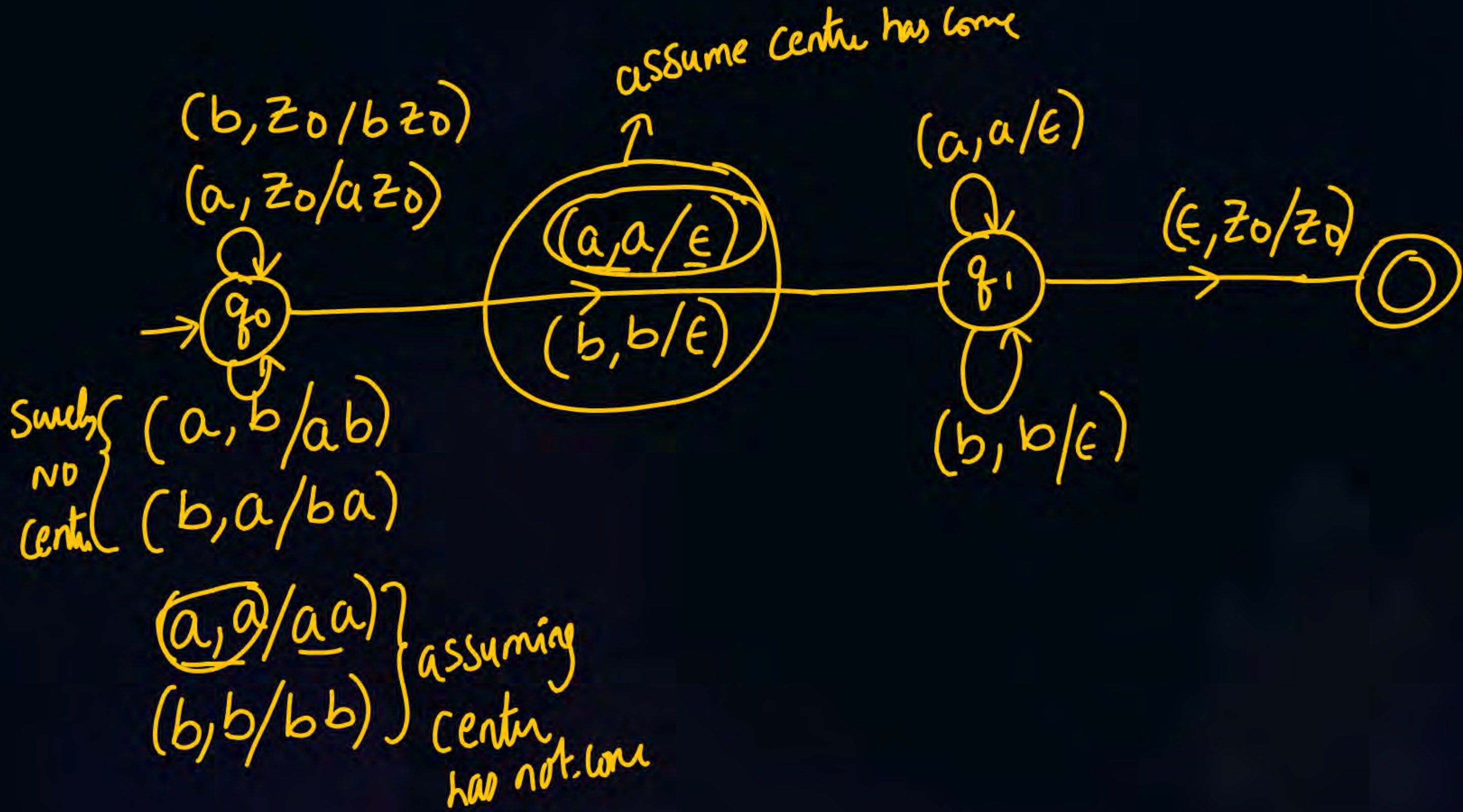
$ww^R / w \in (a+b)^+ \rightarrow$ even length palindromes

$abba$ $baab$ $abbbba$ $baaaba$
 ↑ ↑ ↑ ↑

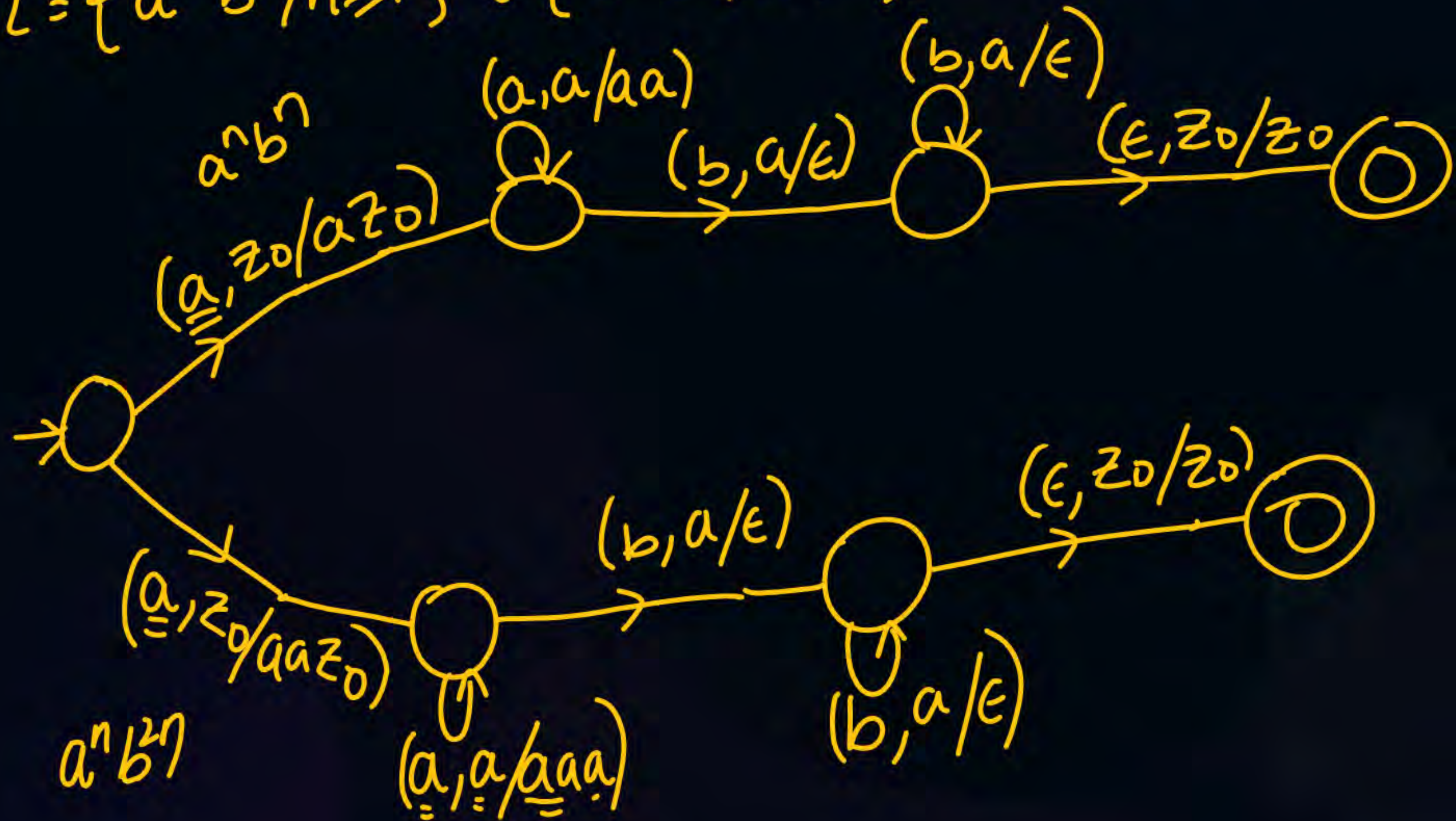
centre is always present b/w two a's & 2 b's

$\check{a} \check{a} \check{a} \check{a}$
 ↑ ↑ ↑
 may may may

$a b b b b a$
 ↑ ↑ ↑ ↑ ↑
 x x



$$L = \{a^n b^n / n \geq 1\} \cup \{a^n b^{2n} / n \geq 1\} \rightarrow \text{NPD A}$$





We cannot convert NPDA to DPDA
because NPDA is more powerful than DPDA

$a^{m+n} b^n c^m$ $\rightarrow$ XReg $\sqrt{DCFL}$ $\sqrt{CFL}$ $\sqrt{CSL}$

$a^m b^{m+n} c^n / n, m \geq 1$

$\begin{array}{cccc} \underline{a^m} & \underline{b^m} & \underline{b^n} & \underline{c^n} \\ \text{Push} & \text{Pop} & \text{Push} & \text{Pop} \end{array}$
 $\rightarrow$ $DCFL$ ✓
 = Reg X



THANK - YOU